Airbnb Price Prediction Using Linear Regression

Project Objective

The main objective of this project is to predict Airbnb listing prices using machine learning based on important property-related features such as bedrooms, bathrooms, room type, availability, and review ratings.

Dataset Used

The dataset contains Airbnb listing information with features including price, accommodates, bathrooms, bedrooms, beds, room_type, property_type, minimum_nights, number_of_reviews, review_scores_rating, availability_365, and instant_bookable.

Data Preprocessing

Missing values were handled carefully. Numerical missing values were filled using median values, categorical features were converted into numerical format, feature scaling was applied using StandardScaler.

Model Used

A Linear Regression model was selected because it is simple, interpretable, and suitable for predicting continuous numerical values such as price.

Model Training Process

The dataset was divided into training and testing sets using an 80:20 split. The model was trained on the training data and predictions were generated on the test data.

Performance Metrics

MAE: 31.00

RMSE: 59.59

R^2 Score: 0.63

Interpretation of Results

The R^2 score of 0.63 indicates that the model explains approximately 63% of the variation in Airbnb prices, showing moderate prediction capability.

Visualization

An Actual vs Predicted Price scatter plot was created to compare predicted values with actual prices. Most points are close to the ideal diagonal line, indicating reasonable prediction accuracy.

Conclusion

This project demonstrates how machine learning can be applied to real-world pricing problems. Linear Regression provided a clear baseline model for predicting Airbnb prices and can be improved further using advanced algorithms.